

1.3 Primes and Unique Factorization

Goal: We want to show that every integer is a product of primes, that this factorization is unique in some sense.

Example. Find a prime factorization of 12. $= 2 \cdot 2 \cdot 3$
 $= (-2)(-2) \cdot 3$
 $= 2(-2) \cdot (-3)$

Definition. An integer p is prime if $p \notin \{0, \pm 1\}$, and the only divisors of p are ± 1 and $\pm p$.

Let $\mathbb{P}$ be the set of all primes.

$$d(p) = \{\pm 1, \pm p\}$$

Remark: In many other sources, prime numbers are defined to solely include positive integers. However we will stick to the textbook's convention.

Examples of primes

$2, -2, \pm 3, \pm 5, \pm 7, \pm 13, 839, 2473$
 1229

Facts about primes and divisibility.

- There are infinitely many distinct primes (Group Exercise).
- Let p and q be prime integers. Then $p \mid q$ if and only if $p = \pm q$.

Theorem 1.5. Let $b, c \in \mathbb{Z}$, and let p be an integer with $p \notin \{0, \pm 1\}$. Then p is prime if and only if p satisfies the condition:

$$\text{if } p \mid bc, \text{ then } p \mid b \text{ or } p \mid c. \quad \star$$

Recall Theorem 1.4, If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.

Proof. ($\Rightarrow$): Assume $p \in \mathbb{P}$, we want to show p has property $\star$

suppose $p \mid bc$, note that $\gcd(p, b)$ is either 1 or p
 If $\gcd(p, b) = 1$, by theorem 1.4 $p \mid c$.

If $\gcd(p, b) = p$ then $p \mid b$ in either case we are done & p has property $\star$.

($\Leftarrow$): Suppose that p satisfies $p \mid bc$ implies $p \mid b$ or $p \mid c$.

let s be a divisor of p , so there exists $t \in \mathbb{Z}$ such that $s \cdot t = p$. Since $p \mid p$ then $p \mid s$ or $p \mid t$.

If $p \mid s$, since $s \mid p$ then $p = \pm s$.

If $p \mid t$, t is also a divisor of p so $p = \pm t \Rightarrow s = \pm 1$ $\square$

Since s was chosen arbitrarily from $d(p)$ and s

is either $\pm p$ or ± 1 then $d(p) = \{\pm p, \pm 1\}$ hence $p \in \mathcal{P}$. $\square$

Remark. Because Theorem 1.5 is biconditional, we've just given a **new** definition of what it means to be prime.

Corollary 1.6. If $a_1, a_2, \dots, a_n \in \mathbb{Z}$, p is prime, and $p \mid a_1 a_2 \cdots a_n$, then p divides at least one of the a_i .

Proof. Sketch: $\hookrightarrow$ If $p \mid a_1(a_2 \cdots a_n)$ by the theorem

$$p \mid a_1 \text{ or } p \mid (a_2 \cdots a_n)$$

if $p \mid a_1$ we are done else, by theorem

$$p \mid a_2 \text{ or } p \mid a_3 \cdots a_n$$

if $p \mid a_2$ we are done else, by theorem

$$p \mid a_3 \text{ or } p \mid a_4 \cdots a_n \dots$$

$\vdots$

$$p \mid a_{n-1} a_n \text{ then } p \mid a_{n-1} \text{ or } p \mid a_n.$$

Theorem 1.7. Every integer $n \notin \{0, \pm 1\}$ is a product of primes.

Proof. Let $S = \{x \mid x > 0, x \in \mathbb{Z}, x \text{ is not a product of primes}\}$

For the sake of contradiction say $S \neq \emptyset$.

By the well ordering axiom since $S \subseteq \mathbb{N}_0$ & $S \neq \emptyset$ there exists a smallest element t .

Thus t is not the product of primes, in particular it is not a prime.

Thus exist at least one divisor m

$$m \cdot n = t \text{ some } n. \text{ note } m < t \text{ by}$$

the minimality of t , both $m, n \notin S$ & thus products of primes

Remark: An integer other than $0, \pm 1$ that is not prime is called a composite

$$m = p_1 \cdots p_\ell$$

$$n = p_{\ell+1} \cdots p_{\ell+s}$$

where $p_i \in P$ for $i \in \{1, \dots, \ell+s\}$

thus $t = p_1 \cdots p_\ell p_{\ell+1} \cdots p_{\ell+s}$

this is a contradiction on
 $t \in S$.

Thus $S = \emptyset$. and every integer
not $0, \pm 1$, is a prod of
primes.

Remark. This theorem gives the *existence* of a prime factorization for every integer $n \neq 0, \pm 1$. Theorem 1.8 gives the (almost) uniqueness of this factorization.

Theorem 1.8. [Fundamental Theorem of Arithmetic] (FTA) Every integer $n \notin \{0, \pm 1\}$ is a product of primes, and these primes are unique in the following sense:
 If $n = p_1 p_2 \dots p_r$ and $n = q_1 \dots q_s$ for some primes p_i, q_j , then $r = s$. Also for all $i \in \{1, \dots, r\}$ we have $p_i = \pm q_j$ for some $j \in \{1, \dots, r\}$

Proof Strategy: We will use Mathematical Induction.

$$\underline{\text{Ex}} \quad 24 = 2 \cdot 2 \cdot 2 \cdot 3 = 2(-2)(-2)3$$

I **Base Case:** First we show that FTA holds when $r = 2$.

$$n = p_1 p_2 = q_1 \dots q_s$$

II **Inductive Step:** Next we assume FTA holds for $r - 1$ and show that this implies FTA for r

Notice then that by I the theorem holds for $r = 2$. By II the theorem holds for $r = 3$. By II the theorem holds for $r = 4$. By II the theorem holds for $r = 5$ and so on. Therefore FTA will hold for all r .

Proof. Base case i.e. suppose $n = p_1 p_2 = q_1 q_2 \dots q_s$

we want to show that $s = 2$ and $p_i = q_i$ some i .

Since $p_1 p_2 = q_1 q_2 \dots q_s$ then $p_1 \mid q_1 q_2 \dots q_s$

so by corollary 1.6 $p_1 \mid q_j$ for some $j \in \{1, \dots, s\}$

since q_j is a prime then its only divisors are $\pm 1, \pm q_j$
 but $p_1 \neq \pm 1$ therefore $p_1 = \pm q_j$.

Dividing ~~by~~ by p_1 we have $p_2 = q_1 q_2 \dots q_{j-1} q_{j+1} \dots q_s$

again by cor 1.6 $p_2 \mid q_i$ & similarly implies $p_2 = \pm q_i$

$$1 = q_1 q_2 \dots q_{i-1} q_{i+1} \dots q_{j-1} q_{j+1} \dots q_s$$

If $s \neq 2$ then $q_k \mid 1$ which implies $q_k = \pm 1$

but ± 1 is not prime, this is a contradiction

$$\text{so } s = 2$$

II Inductive step: Assume FTA holds for
when $n = p_1 \cdots p_{r-1}$ & show it implies
that FTA holds for $m = p_1 \cdots p_{r-1} p_r$.

suppose $m = p_1 \cdots p_{r-1} p_r = q_1 q_2 \cdots q_s$

then $p_r \mid q_1 q_2 \cdots q_s$ & by cor 1.6

$p_r \mid q_j$ for some j , and as before
this implies $p_r = \pm q_j$

& so $\frac{m}{p_r} = p_1 \cdots p_{r-1} = q_1 q_2 \cdots q_{j-1} q_{j+1} \cdots q_s$

By the inductive assumption

$r-1 = s-1$ and for all i , $p_i = \pm q_k$ for some
 $k \in \{1, \dots, r-1\}$

then $r = s$.

Thus the fundamental Theorem of Arithmetic
holds for all pos integers. $\square$

Group Exercises.

1. (Theorem 1.10) Prove, by contradiction, that if n is an integer greater than 1 and n is not prime, then n is divisible by some prime $p \leq \sqrt{n}$.

By contradiction, say $n \in \mathbb{Z}$ $n > 1$, $n \notin \mathbb{P}$ but none of its prime divisors are less than $\sqrt{n}$.

Then if $n = p_1 \cdots p_l$ then $p_i > \sqrt{n}$, $p_i^2 > n$.

$$n^2 = (p_1 \cdots p_l)(p_1 \cdots p_l)$$

$$= p_1^2 p_2^2 \cdots p_l^2 > n p_2^2 \cdots p_l^2 > n \cdot n p_3^2 \cdots p_l^2 > \cdots > n^l$$

$n^2 > n^l$ this implies $0 < l < 2$ so $l = 1$. so $n = p_1$

2. Show that $\sqrt{2}$ is irrational using the Fundamental Theorem of Arithmetic.

Assume $\sqrt{2}$ is rational, i.e. $\exists a, b \in \mathbb{Z}$

$$\sqrt{2} = \frac{a}{b} \quad \text{where } a \text{ \& } b \text{ have no common divisors}$$

By FTA $a = p_1 \cdots p_n$ $b = q_1 \cdots q_s$

$$2 = \frac{a^2}{b^2} \Rightarrow b^2 \cdot 2 = a^2 \Rightarrow \underbrace{q_1^2 \cdots q_s^2 \cdot 2}_{2s+1 \text{ primes}} = \underbrace{p_1^2 \cdots p_n^2}_{2n \text{ primes}}$$

contradiction
as 2 is not a prime.

3. Show there exists an infinite number of primes. (Hint: Try doing this by contradiction.)

Suppose only a finite number of primes exists.

Take $p_1 p_2 \cdots p_n$ to be the product of all odd prime.

$$p_1 p_2 \cdots p_n + 2 \in \mathbb{Z}$$

$$\exists p \text{ s.t. } p \mid p_1 p_2 \cdots p_n + 2$$

By FTA

$$\underbrace{2s+1}_{\text{odd}} = \underbrace{2n}_{\text{even}}$$

a contradiction
□

$$P^N = P_1 P_2 \dots P_n + 2$$

Start over & let $P_1 \dots P_n$
be the prod of all primes

$$\underbrace{P_1 \dots P_n + 1}$$

is either prod of primes
or a prime

if prime $P_l = P_1 \dots P_n + 1 \geq P_1 \dots P_n >$

$$1 > P_1 \dots P_{i-1} P_{i+1} \dots P_n > 0$$

a contradiction

thus not a prime

else $\exists p$ s.t. $p \mid P_1 \dots P_n + 1$

$$P_l = P_1 \dots P_n + 1$$

$$l = P_1 \dots P_{j-1} P_{j+1} \dots P_n + \frac{1}{p}$$

$$l - p_1 \cdots p_j + p_j + \cdots p_n = \frac{1}{p} \in \mathbb{Z}$$

thus $p = \pm 1$ ~~is~~ contradiction